MICROBIAL POLLUTION CONTROL

Class 10 | Science-2 | Ch-7 Introduction to Microbiology | Pg. 83–85

WHY DO WE NEED MICROBES ?

① Solid waste, sewage & pollutants keep **increasing** with pollution.
② So diseases **increase** & environment gets **degraded**.
③ Densely populated countries like **India** → cities suffer the most.
∴ **Solution → use of MICROBES for disposal & cleaning.**

I. LAND-FILLING SITES

Defⁿ :- ① The method of disposal of degradable waste (सड़ने वाला कचरा)
② which gets accumulated in urban areas,
③ by dumping it in large pits lined with plastic sheets.

Steps :-

1) Large **pits are dug** in open spaces — far away from residential area.
↓
2) Pits are lined with a **PLASTIC SHEET**
→ **Why ?** to stop leaching (रिसना) of toxic & harmful material into the **soil.**
↓
3) **Compressed waste** (दबाया हुआ) is dumped in the pit.
↓
4) Covered with **layers** of → **soil, saw dust, leafy waste & specific biochemicals.**
↓
5) **Bioreactors** are mixed at some places.

LET'S THINK (PG. 83)

① Why are we asked to **segregate** (अलग करना) wet & dry waste at home ?
→ Wet waste is **degradable** → it can be turned into compost / biogas. Dry waste is **non-degradable** → it must be recycled.
② Which is the most appropriate method of disposal of **dry waste** ?
→ **Recycling**; what is left over goes to the **land-filling site.**

6) **Microbes** present in soil & top layers decompose the waste.
↓
7) Completely filled pit is **sealed** with **soil slurry** (गाढ़ा घोल)
↓
8) After a few days → **BEST QUALITY COMPOST** is formed.
↓
9) Compost is removed → **site can be RE-USED.**

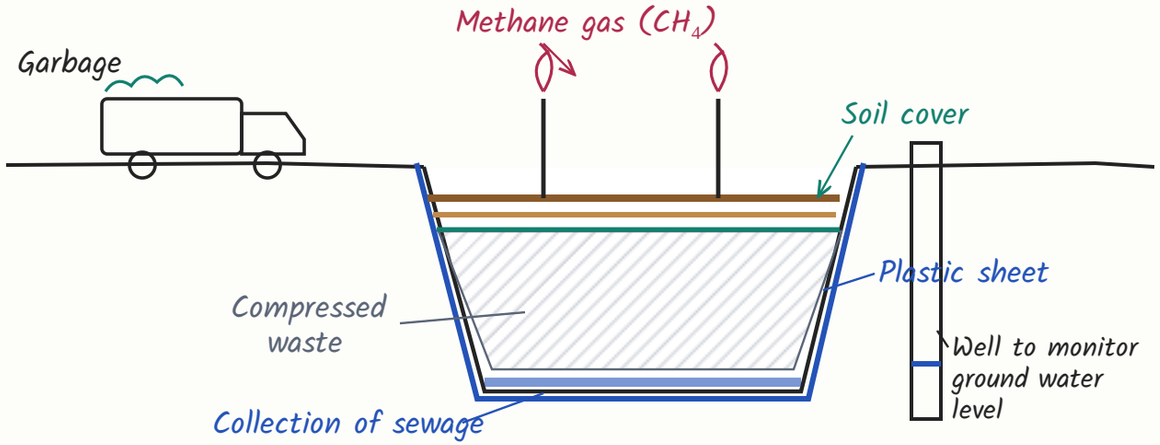


Fig. 7.9 – Modern land-filling site

✦ **ALWAYS REMEMBER** — A land-filling site is **not wasted.** After the compost is removed, the **same pit can be used again.**

OBSERVE (PG. 83) — Garbage vans of the grampanchayat & municipality now **compact** the garbage. **Advantage** → volume of garbage decreases → more waste in one trip → **less fuel, less time & less space** needed in the pit.

2. SEWAGE MANAGEMENT

① **Sewage** (मल-जल) = waste water + waste matter thrown out of houses.

PLACE	WHAT IS DONE WITH THE SEWAGE
In VILLAGES	Disposed off in **nearby soil** OR in a **biogas plant.**
In CITIES	Carried to a **processing unit** → acted upon by **microbial processes.**

Process :-

1) Special **microbes** are mixed with the sewage.
→ These microbes can decompose **any** compound.
↓
2) They do **TWO** jobs :-
① Decompose the carbon compounds of sewage → release $CH_4 + CO_2$
② Destroy the pathogens (रोगकारक जीव) of **cholera, typhoid** etc.
↓
3) **Phenol oxidising bacteria** decompose the xenobiotic chemicals in sewage.

★ **WORD MEANING** **Xenobiotic** = a man-made chemical **foreign to nature** — not made by any living organism. (प्रकृति के लिए पराया रसायन)

Two useful products :-

PRODUCT	USE
SLUDGE (गाद) – settles down	Used again as **FERTILIZER**
WATER – released after treatment	Is **environmentally SAFE.**

④ This whole use of microbes to clean a polluted environment → **BIOREMEDIATION**

3. CLEAN TECHNOLOGY

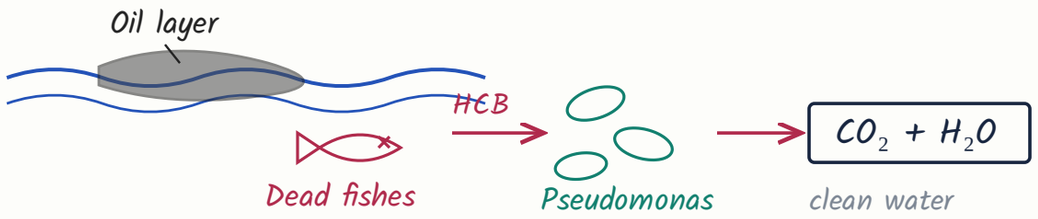
Def^n :- ① Man has made very fast progress in technology,
② but pollution is also increasing at the same speed.
③ Controlling **air, soil & water pollution** with the help of **microbes** is called **CLEAN TECHNOLOGY**.

BASIS OF CLEAN TECHNOLOGY
Microbes have a **natural ability** to decompose man-made chemicals. Hydrocarbons & other chemicals are **transformed** using this ability.

How microbes clean — Part 1

NO	POLLUTION PROBLEM	MICROBE USED	ACTION
1	**Sulphur** present in **fuels**	**Some microbes**	Remove the sulphur from the fuel.
2	Metals — **Cu, Fe, U, Zn** — leach into the environment from **low quality metalloids**	**Thiobacilli Sulphobacilli**	Convert these metals into **compounds** before they leach out.
3	**Oil spill** in the ocean — oil layer is fatal & toxic to aquatic organisms & cannot be removed by machines	**Pseudomonas spp. Alcanovorax borkumensis**	Destroy pyridines & other chemicals. Decompose hydrocarbons → C reacts with O_2 → $CO_2 + H_2O$

✦ **REMEMBER THE NAME**
Bacteria that eat up hydrocarbons are called **HCB** = **H**ydro**c**arbono**c**lastic **B**acteria.
→ **Used to clear oil spills of rivers & oceans.**



3. CLEAN TECHNOLOGY — PART 2

NO	POLLUTION PROBLEM	MICROBE USED	ACTION
4	**Plastic bottles** made of PET (Polyethylene Terephthalate Polyester) Most of the urban garbage today = plastic.	**Vibrio Ideonella sakaiensis**	Decompose the **PET.**
5	**Rubber** present in garbage	**Actinomycetes Streptomyces Nocardia Actinoplanes**	Decompose the **rubber.**
6	**Sulphuric acid (H_2SO_4)** present in acid rain & in material coming out of mines. → It erodes metals of statues, bridges & buildings + pollutes the soil.	**Acidophillium spp. Acidobacillus ferroxidens**	For them H_2SO_4 is a **source of energy** → so they **use it up** & control soil pollution.
7	Water-soluble **salts of Uranium** in – waste of electroplating – effluent of atomic energy plant	**Geobacter**	Convert **soluble** U-salts into **insoluble** salts → so they cannot mix with ground water.

✦ **ALWAYS REMEMBER (PG. 85)**
The plastic used for storing garbage should be **biodegradable POLYLACTIC ACID.** Such material should be used **as per need only** — and the environment should be saved.

MICROBES AND FARMING

4. MICROBIAL INOCULANTS

Defⁿ :- ① Preparations which contain some microbes,
② produced by the process of **FERMENTATION**,
③ which are **sprayed on seeds before sowing OR** released into the plants.

→ Inoculant = मिलाया जाने वाला जीवाणु-घोल | Sowing = बुवाई

What do they do ?

1) Microbes in the inoculant **supply nutrients** to the plant.
2) ∴ They **help in plant growth.**
3) They **improve the quality** of vegetarian food.

e.g Solution of **Azotobacter** + **artificial nitrogenase** → used in **ORGANIC FARMING.**

✓ **BIGGEST ADVANTAGE**
Soil pollution caused by **chemical fertilizers is prevented** by using these solutions.

Problem of chemical pesticides :-

1) Chemical pesticides are used in agriculture.
↓
2) **Fluoroacetamide**-like chemicals get mixed with the **soil.**
↓
3) Harmful effects → ① harmful to **other plants & animals**
② cause **skin diseases** in humans.
↓
4) **These pesticides in the soil can be DESTROYED with the help of microbes.** ✓

✦ **MICROBES HELP THE FARMER IN 2 WAYS**
① **Microbial inoculants** → **FEED** the plant (nutrients).
② **Bio-insecticides** → **PROTECT** the plant (kill the pest). → next board.

5. BIO-INSECTICIDES

(also called **bio-pesticides**) Pest = कीट / फसल का दुश्मन

Def^{n} :- ① **Bacterial & fungal toxins**
② which can destroy pests & pathogens,
③ and are **integrated directly into the plants** with the help of **BIOTECHNOLOGY.**

How does it work ?

1) Toxin gene is put **inside the plant.**
↓
2) The plant itself becomes **toxic to insects.**
↓
3) ∴ Insects **do not consume the plant.**
↓
4) **Crop is saved — without spraying chemical poison. ✓**

④ Like bacteria, some species of **FUNGI** & **VIRUSES** are also useful as pesticides.

e.g **SPINOSAD** — a **by-product of fermentation**, it is a **biopesticide.**

↪ **LINK WITH CH-8 (CELL BIOLOGY & BIOTECHNOLOGY)**
Bt Cotton & Bt Brinjal — a gene from the bacterium **Bacillus thuringiensis** is integrated with the plant's genome. The toxin formed in the leaves & bolls **destroys the alimentary canal of the bollworm** and the pest dies. → **This is a bio-insecticide in action.**

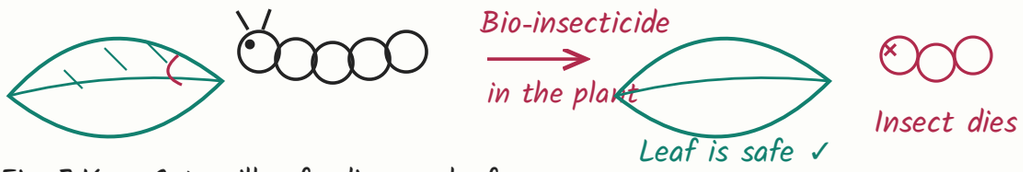


Fig. 7.14 – Caterpillar feeding on leaf

CHEMICAL PESTICIDE V/S BIO-INSECTICIDE

POINT	CHEMICAL PESTICIDE	BIO-INSECTICIDE
Source	Man-made chemical	Toxin of **bacteria / fungi**
How used	**Sprayed** on the crop	**Built into the plant** by biotechnology
Effect on soil	Fluoroacetamide-like chemicals **pollute the soil**	No such chemical residue
Other plants & animals	**Harmful**	Toxic **only to the insect**
Effect on man	Causes **skin diseases**	Safe
Used in	Ordinary farming	**Organic farming**

✦ **ONE-LINE REVISION OF THE WHOLE TOPIC**
① **Land-filling** → solid waste → compost.
② **Sewage management** → dirty water → sludge (fertilizer) + safe water.
③ **Clean technology** → oil, plastic, rubber, acid, uranium → harmless.
④ **Microbial inoculants** → nutrients to plants, no chemical fertilizer.
⑤ **Bio-insecticides** → kill the pest, save the soil.

Board Questions — Exercise, Pg. 86

QUESTION	ANSWER IS ON
3-b. How can the oil spills of rivers & oceans be cleaned ?	**Board 2 – HCB**
3-c. How can the soil polluted by acid rain be made fertile again ?	**Board 3 – No. 6**
3-d. Explain the importance of bio-pesticides in organic farming.	**Board 4**
3-g. Which precautions are necessary for proper decomposition of domestic waste ?	**Board 1**
3-h. Why is it necessary to ban the use of plastic bags ?	**Board 3 – No. 4**

HOME WORK — Draw Fig. 7.9 (modern land-filling site) with labels & write the **7 steps** of land-filling in your own words.